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Estimating Thermal Conductivity from Transient Thermocouple Data Using Physics-Informed Neural Networks

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Abstract

Estimating thermal conductivity from temperature measurements collected by a small number of thermocouples offers an alternative to standardized measurement methods. This study investigates whether an inverse physics-informed neural network (PINN) can provide estimates comparable to a published nonlinear least-squares method (baseline). The PINN was applied to experimental temperature data from an Al6061-T6 aluminum rod in the open dataset by Tomanek and Stutts, estimating thermal conductivity k, convective heat transfer coefficient h, and heating power P. Its performance was compared with the baseline using temperature root mean square error (RMSE) and differences in the estimated parameters. For one experiment and one random seed, the PINN achieved an RMSE of 0.381 °C, compared with 0.377 °C for the baseline, while its estimates of k, h, and P were approximately 13–16% lower. These preliminary results show comparable temperature prediction errors but differences in parameter estimates; further experiments and training runs are needed to assess parameter stability and the reliability of the estimates.

Keywords: physics-informed neural networks; inverse problem; heat equation; thermal conductivity; parameter estimation; partial differential equations

1. Introduction

Thermal conductivity is an important material property for the design and evaluation of materials and thermal systems.Its measurement using standardized methods may require specialized equipment [1]. As an alternative, Tomanek and Stutts proposed estimating thermal properties from temperature measurements collected by a small number of thermocouples [1,2]. This approach provides a way to estimate material properties using experimental temperature data.

Their method estimates the unknown parameters by applying nonlinear least squares to an analytical solution of the heat equation. According to [2], the reported thermal conductivity estimates were within 10% of published nominal values. However, this approach relies on an analytical solution and an appropriate initialization of the parameters [1]. This raises the question of whether similar estimates can be obtained using a method that incorporates the governing equation directly, without explicitly relying on its closed-form solution.

Physics-informed neural networks (PINNs) are neural networks trained to fit observed data while incorporating the constraints imposed by physical laws, such as partial differential equations (PDEs) [3]. In inverse problems, unknown physical parameters can be optimized jointly with the network parameters. PINNs have also been investigated for heat transfer problems, including problems with noisy data and partially unknown physics [4].

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This applied research investigates whether an inverse PINN can estimate the thermal conductivity of an aluminum rod using experimental temperature measurements from a small number of thermocouples. The PINN is compared with the nonlinear least-squares baseline using the root mean square error (RMSE) of the predicted temperatures and the relative difference in estimated thermal conductivity. The contribution of this study is a preliminary experimental evaluation of an inverse PINN using an open dataset of temperature measurements. The preliminary results show comparable temperature prediction errors for both methods, although their estimated physical parameters differ.

The remainder of this paper is organized as follows. Section 2 describes the experimental data, the physical model, and the estimation methods. Section 3 presents the preliminary results, and Section 4 discusses their interpretation, limitations, and possible directions for future work.

2. Materials and Methods

2.1. Dataset

The data used in this study were collected by Tomanek and Stutts and released as an open dataset [5], described in [1].

The dataset provides raw temperature measurements from a thin rod modeled with the one-dimensional heat equation. The experimental setup features a rod heated at one end by a nichrome wire (34 gauge), with six thermocouples positioned along its length. The rod is placed inside a wind tunnel, and a data acquisition system—based on an Arduino board—is used to record measurements and control the heating power (via 10-bit PWM). The temperatures are reported as excess temperature, $\theta = T - T_\infty$, where $T_\infty = 20.5\,^\circ\mathrm{C}$ is the ambient temperature. The thermocouples are labeled TC1 to TC6, ordered from the closest to the farthest from the heater.

The raw data are stored in `.xls` files, each corresponding to a single test. Each file name encodes the boundary temperature, the material, and the test number. For example, `50al1` corresponds to aluminum, a boundary temperature of 50,°C, and test 1. Measurements were sampled every 0.2 s over approximately 300 s. Each file contains the time, the excess temperature registered by each thermocouple, and the power setting on a scale from 0 to 1023.

For this preliminary study, we used three files of the aluminum alloy Al6061-T6 rod, obtained at a constant wind speed of 5 m/s: `50al1`, `75al1`, and `100al1`, with boundary temperatures of 50, 75, and 100 °C, respectively. The dataset also includes two additional subsets, one with varying wind speed and another with rods produced by additive manufacturing, which are not covered in this work.

The dataset is distributed under a CC BY 4.0 license. The only modification we made was converting the files from `.xls` to `.csv`.

2.2. Physical Model

The inverse PINN aims to represent the 1D heat equation. As explained in Section 2.1, the study involves a rod modeled as a one-dimensional thin fin with convective losses. The objective of this representation is to work with a physical model to estimate the parameters k, h and P (k for this preliminar work) and support it with mathematical rigor.

2.2.1. The Heat Equation 1D

Note that the geometry of rod is simplified in order to apply the 1D heat equation to a more familiar system. The excess temperature $\theta(x,t)$ satisfies

$$\frac{\partial \theta}{\partial t} = \kappa \frac{\partial^2 \theta}{\partial x^2} - \nu\,\theta, \qquad \kappa = \frac{k}{\rho c}, \quad \nu = \frac{h\,s}{\rho A c}, \tag{1}$$

where x is the distance from the heated end (m), t is time (s), k is the thermal conductivity ($\mathrm{W\,m^{-1}K^{-1}}$), ρ is the density ($\mathrm{kg\,m^{-3}}$), c is the specific heat ($\mathrm{J\,kg^{-1}K^{-1}}$), h is the convective heat transfer coefficient ($\mathrm{W\,m^{-2}K^{-1}}$), $s = \pi D$ is the perimeter and $A = \pi D^2/4$ is the cross-sectional area of a rod of diameter D.

On the left side of Eq. (1), $\partial\theta/\partial t$ is the rate at which the excess temperature changes with time at a fixed point of the rod.

The first term on the right side contains the thermal diffusivity $\kappa = k/(\rho c)$, which depends on the material properties k, ρ and c. It multiplies the second derivative of the temperature with respect to x, which accounts for the curvature of the temperature profile along the rod; together, they describe heat conduction within the rod.

To describe convection, we introduce $\nu = hs/(\rho A c)$, which groups the parameters h and s in the numerator and ρ, A and c in the denominator. Multiplying ν by θ gives the term that accounts for the heat lost from the lateral surface of the rod to the air. This term has a negative sign because it removes heat, and it grows as the rod becomes hotter than its surroundings.

Combining the three terms, the change in temperature results from the heat gained by conduction and the heat lost by convection.

The heated end receives a heat input and the far end is insulated:

$$-kA\left.\frac{\partial\theta}{\partial x}\right|_{x=0} = P\left(1 - e^{-\alpha(t+\tau)}\right), \qquad \left.\frac{\partial\theta}{\partial x}\right|_{x=L} = 0, \tag{2}$$

where P is the steady-state power entering the rod (W), α and τ are the rise rate ($\mathrm{s^{-1}}$) and the delay (s) of the heating, and L is the rod length (m). [Frase: what each condition means physically.]

To represent the initial condition, we fix an instant $t = 0$ at which the heater is not yet turned on. At this instant the rod is in thermal equilibrium with the surrounding air, so $T = T_\infty$ and the excess temperature is zero,

$$\theta(x, 0) = 0. \tag{3}$$

This initial condition is necessary to obtain a unique solution of Eq. (1).

2.2.2. Known Parameters

The known parameters used to develop the inverse PINN were taken from the code released by the authors of the dataset [6]. They include the material properties (density and specific heat), the dimensions of the rod (diameter and length), and the test conditions (ambient temperature and wind speed). These are nominal values, not measured in this work, and are listed in Table 1.

Table 1 lists the values used for the aluminum rod.

Table 1. Known parameters of the Al6061-T6 rod and the test conditions [6].

Parameter	Value	Unit
Density, ρ	2767.99	$\mathrm{kg\,m^{-3}}$
Specific heat, c	896	$\mathrm{J\,kg^{-1}K^{-1}}$
Diameter, D	3.175	mm
Length, L	150	mm
Ambient temperature, T_∞	20.5	°C
Wind speed, v	5	$\mathrm{m\,s^{-1}}$
Thermocouple positions, x_i	0, 10, 20, 30, 40, 50	mm

The perimeter s and the cross-sectional area A were computed from the rod diameter D using the formulas defined above. The thermocouple positions are also known: the six sensors are spaced 10 mm apart, from $x = 0$ at the heated end to $x = 50$ mm. Since the rod is 150 mm long, the measurements cover only the first third of its length, and the remaining part, including the insulated end at $x = L$, is not measured.

2.2.3. Estimated Parameters

The inverse PINN estimates three unknown parameters of the physical model: the thermal conductivity k, the convective heat transfer coefficient h and the steady-state heating power P (Eqs. (1) and (2)). Table 2 summarizes them.

Table 2. Parameters estimated by the inverse PINN and their initial values.

Parameter	Meaning	Unit	Initial value
k	Thermal conductivity	$\mathrm{W\,m^{-1}K^{-1}}$	100
h	Convective heat transfer coefficient	$\mathrm{W\,m^{-2}K^{-1}}$	130.8
P	Steady-state heating power	W	1.0

The initial value of h was computed with the Churchill–Bernstein correlation for crossflow over a cylinder [7], as in [1], using the wind speed and the rod diameter, which gives $h = 130.8\ \mathrm{W\,m^{-2}K^{-1}}$. The parameters k and P were initialized with the generic values $100\ \mathrm{W\,m^{-1}K^{-1}}$ and 1.0 W, respectively; no result from the baseline was used for initialization.

The heating parameters α and τ are not estimated by the network. They are fixed beforehand for each test: the temperature gradient at $x = 0$ is approximated with a second-order forward finite difference using the first three thermocouples (TC1–TC3), and the exponential rise in Eq. (2) is fitted to it by least squares. Here α determines the rate at which the heat input rises and τ is the heating delay.

The same values are used in the baseline and in the PINN, so that the two methods differ only in how k, h and P are estimated.

2.3. Baseline

The baseline is the nonlinear least-squares estimator proposed by Tomanek and Stutts [2], whose code is publicly available [6]. It estimates unknown parameters by minimizing the squared error between the model predictions and the measured temperatures. The model is the closed-form solution of Eq. (1) with the boundary conditions of Eq. (2), written as a Fourier series with 100 terms. The authors minimize the error with the Levenberg–Marquardt algorithm [8], an iterative method for nonlinear least squares, through the function `scipy.optimize.least_squares`.

The estimation proceeds in four steps: (i) α and τ are fitted to the temperature gradient at $x = 0$; (ii) a steady-state fit using the last 20 samples of each thermocouple gives the ratio h/k; (iii) the initial values of h, k and P are computed from the previous steps and from the Churchill–Bernstein correlation; and (iv) h, k and P are estimated by fitting the closed-form solution to all the thermocouple data.

For this preliminary study, we ran the authors' code with the parameters of the aluminum rod on the files `50al1`, `75al1` and `100al1`. Unlike the original implementation, our run does not compute the uncertainty of the estimated parameters, which the authors obtain from the Jacobian matrix.

2.4. Inverse PINN

The inverse problem is addressed using a physics-informed neural network (PINN) [3, 4] that approximates the excess temperature as a function of position and time, $\hat{\theta}(x,t)$. The network uses experimental temperature measurements while incorporating the governing heat equation (Eq. (1)) and its boundary and initial conditions (Eqs. (2) and (3)). The thermal conductivity k, convective heat transfer coefficient h, and heating power P are treated as trainable parameters; they are trained in logarithmic scale to keep them positive.

The network is trained by minimizing the following loss function:

$$\mathcal{L} = \lambda_d \mathcal{L}_{\text{data}} + \lambda_f \mathcal{L}_{\text{PDE}} + \lambda_0 \mathcal{L}_{x=0} + \lambda_L \mathcal{L}_{x=L} + \lambda_i \mathcal{L}_{\text{IC}}, \tag{4}$$

where $\mathcal{L}_{\text{data}}$ measures the difference between predicted and measured temperatures, $\mathcal{L}_{\text{PDE}}$ penalizes violations of the heat equation at interior collocation points, and $\mathcal{L}_{x=0}$ and $\mathcal{L}_{x=L}$ enforce the boundary conditions at the heated and insulated ends, respectively. The term $\mathcal{L}_{\text{IC}}$ enforces the initial condition $\theta(x,0) = 0$. Each term is a mean squared error computed with dimensionless variables. The coefficients λ_d, λ_f, λ_0, λ_L, and λ_i control the relative contributions of these terms.

The loss weights are set to $\lambda_d = 10$, $\lambda_f = 1$, $\lambda_0 = 1$, $\lambda_L = 1$, and $\lambda_i = 10$. The network parameters and the unknown physical parameters are optimized jointly with the Adam optimizer [9], using one of every five time samples of the six thermocouples. The remaining settings are listed in Table 3.

Table 3. Configuration of the inverse PINN.

Setting	Value
Architecture	3 hidden layers, 64 neurons, tanh
Inputs / output	$(x/L,\ t/300\,\text{s})$ / $\theta/40\,^{\circ}\text{C}$
Optimizer	Adam
Learning rate (network / parameters)	10^{-3} / 10^{-2}
Iterations	40 000
Collocation points per iteration	2000 (PDE), 300 ($x = 0$ and $x = L$), 300 (IC)
Training data	every 5th sample (1 s) of TC1–TC6
Initial values of k, h, P	100, 130.8, 1.0
Random seed	0

2.5. Metrics

The inverse PINN is evaluated using the relative difference in thermal conductivity with respect to a reference value and the root mean square error (RMSE) of the predicted temperatures. The metrics are defined as

$$\varepsilon_k = \frac{|k_{\text{PINN}} - k_{\text{ref}}|}{|k_{\text{ref}}|} \times 100\%, \qquad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{\theta}_i - \theta_i)^2}, \tag{5}$$

where k_{PINN} is the thermal conductivity estimated by the PINN, k_{ref} is the reference value, $\hat{\theta}_i$ and θ_i are the predicted and measured excess temperatures, respectively, and N is the number of evaluated observations (all time samples of the six thermocouples). In this preliminary study, k_{ref} is the value estimated by the baseline (Section 2.3), which is itself an estimate and not a ground truth. The relative error in k is the metric used by Tomanek and Stutts to compare their estimates with nominal values [2], whereas the RMSE measures the typical magnitude of temperature prediction errors in the same units as the measurements. Both metrics are computed for the baseline and for the PINN.

2.6. Software

The project was implemented in Python using NumPy for numerical computations, pandas for loading and processing the experimental data, and Matplotlib for visualizing temperature measurements and model predictions. PyTorch [10] was used to implement the inverse PINN and compute the derivatives required by the governing heat equation through automatic differentiation.

For the baseline, we reimplemented in Python the estimation procedure of Tomanek and Stutts [6], which relies on SciPy [11] (`scipy.optimize.least_squares`) as described in Section 2.3. The experimental data were converted from Excel to CSV format for processing. The code and instructions to reproduce the results are available at https://github.com/queladrian/Artificial-Intelligence.

3. Preliminary Results

3.1. Baseline

The baseline nonlinear least-squares method was used to estimate the thermal conductivity k and convective heat transfer coefficient h from the three aluminum rod tests, `50al1`, `75al1`, and `100al1`. Table 4 summarizes the estimated parameters and the root mean square error (RMSE) for each test.

The estimated thermal conductivity values range from 146.5 to 153.7 $\mathrm{W\,m^{-1}K^{-1}}$, showing consistency across the three tests despite the different boundary temperatures. The RMSE increases from 0.38 °C for `50al1` to 1.07 °C for `100al1`. The cause of this increase was not investigated in this preliminary study.

Table 4. Baseline estimates for the aluminum rod at a wind speed of 5 m/s.

Test	k ($\mathrm{W\,m^{-1}K^{-1}}$)	h ($\mathrm{W\,m^{-2}K^{-1}}$)	RMSE (°C)
`50al1`	146.5	141.5	0.38
`75al1`	153.7	139.1	0.67
`100al1`	148.7	139.0	1.07

3.2. Inverse PINN

The inverse PINN was trained on the `50al1` dataset only. Table 5 compares its estimates after 40,000 iterations with those obtained using the baseline method on the same test.

Table 5. Baseline and inverse PINN results for `50al1`.

Method	k	h	P (W)	RMSE (°C)
Baseline	146.5	141.5	1.42	0.377
PINN (40,000 iterations)	126.8	119.1	1.20	0.381

The two methods achieve similar RMSE values, with 0.381 °C for the PINN and 0.377 °C for the baseline. Thus, for this test, the PINN reproduces the measured temperatures with a comparable level of error.

However, the estimated physical parameters differ between the methods. Relative to the baseline estimates, the PINN estimates are approximately 13.45% lower for k, 15.83% lower for h, and 15.49% lower for P. These results show that similar temperature prediction errors do not necessarily imply similar parameter estimates.

These findings are preliminary because the PINN was evaluated on only one test using a single random seed. Furthermore, the estimated parameters were still changing at the end of the 40,000 training iterations, with k increasing by approximately 0.5 $\mathrm{W\,m^{-1}K^{-1}}$

per additional 1,000 iterations. An earlier run stopped after 15,000 iterations yielded $k = 114.5\ \mathrm{W\,m^{-1}K^{-1}}$ and an RMSE of 0.558 °C. Therefore, the estimates reported here should be considered provisional rather than converged results.

4. Discussion

The preliminary results show that the inverse PINN achieved a temperature prediction error similar to that of the baseline for the `50a11` test, as shown in Table 5. However, the estimated values of k, h, and P were approximately 13–16% lower than the baseline estimates. These parameters were also still changing when training was stopped, so the results should be considered preliminary.

Several factors could explain the differences between the estimated parameters. First, the PINN may not have fully converged, since the parameter estimates were still changing during training. Second, the parameters may be difficult to identify independently, meaning that different combinations could produce similar temperature predictions. Finally, the two methods used different initial parameter values. These explanations remain hypotheses and require further testing.

This study has several limitations. The PINN was evaluated on only one experiment and with one random seed. The baseline estimate is not the true material property, and the estimated conductivity has not yet been compared with a reference value for the material. In addition, the thermocouples cover only part of the rod, while α and τ were fixed using the baseline fit. The baseline implementation also does not provide estimates of parameter uncertainty.

Future work should first investigate the parameter differences by using longer training runs, different initial values, and multiple random seeds. The study could then be extended to the other aluminum experiments and to datasets with different wind speeds and manufacturing methods. Comparing the estimates with reference material properties and testing alternative models, including a neural network without the physics-based loss, would also help evaluate the contribution of the physical constraints.

References

1. Tomanek, L.B.; Stutts, D.S. Data on the validation to determine the material thermal properties estimation via a one-dimensional transient convection model. *Data in Brief* **2022**, *40*, 107632. https://doi.org/10.1016/j.dib.2021.107632.
2. Tomanek, L.B.; Stutts, D.S. Material thermal properties estimation via a one-dimensional transient convection model. *Applied Thermal Engineering* **2021**, *184*. https://doi.org/10.1016/j.applthermaleng.2020.116362.
3. Raissi, M.; Perdikaris, P.; Karniadakis, G.E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* **2019**, *378*, 686–707. https://doi.org/10.1016/j.jcp.2018.10.045.
4. Cai, S.; Wang, Z.; Wang, S.; Perdikaris, P.; Karniadakis, G.E. Physics-informed neural networks for heat transfer problems. *Journal of Heat Transfer* **2021**, *143*.
5. Tomanek, L.; Stutts, D. Data on the validation to determine the material thermal properties estimation via a one-dimensional transient convection model. Mendeley Data, V1, 2021. https://doi.org/10.17632/sbcf36976g.1.
6. Tomanek, L.B.; Stutts, D.S. oneDkhEstimator. Zenodo, 2021. https://doi.org/10.5281/zenodo.5683312.
7. Churchill, S.W.; Bernstein, M. A correlating equation for forced convection from gases and liquids to a circular cylinder in crossflow. *Journal of Heat Transfer* **1977**, *99*, 300. https://doi.org/10.1115/1.3450685.

8. Marquardt, D.W. An algorithm for least-squares estimation of nonlinear parameters. *Journal of the Society for Industrial and Applied Mathematics* **1963**, *11*, 431–441. https://doi.org/10.1137/0111030.
9. Kingma, D.P.; Ba, J. Adam: A method for stochastic optimization. In Proceedings of the International Conference on Learning Representations (ICLR), 2015.
10. Paszke, A.; et al. PyTorch: An imperative style, high-performance deep learning library. In Proceedings of the Advances in Neural Information Processing Systems 32, 2019.
11. Virtanen, P.; et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* **2020**, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2.